

GENERAL INTRODUCTION:

Presents the project context (urban transport challenges in Algeria), identifies specific problems (lack of info, overcrowding, poor organization), and states the main objectives (smart displays, surveillance, passenger counting, route optimization).

THESIS STRUCTURE:

Briefly explain how the thesis is organized and what each chapter will cover.

CHAPTER 1: Problem Statement and Overview

- **Background of the Study:** Transport situation in Algeria (growth, challenges, statistics)
- **Project Context**
- **Problem statement**
- **Objectives**
- **Systems engineering concepts and approaches:** Introduce system design thinking, lifecycle, and modeling approaches.
- **IoT introduction.**
- **Introduction to machine learning and AI**

CHAPTER 2: State of the Art and Technical Context

- **Background on Tracking Systems:** Overview of how modern tracking works
- **Market Study & Existing Analysis:**
- **User Survey & Statistics:** Data from your survey (charts/graphs).
- **Comparative Study:** Similar works and the "Gap" analysis table.
- **Project Impact:** The social/economic value of your solution

CHAPTER 3: System Analysis and Design

- **Stakeholders:** students, bus drivers, operators, administrators.
- **Functional Requirements**
- **Non-Functional Requirements**

- **UML Diagrams:** Use case diagrams, class/data model diagrams, sequence/activity diagrams.
- **UX Design:** Wireframes, mockups, and user interface considerations.
- **Architecture Design:** Overall system architecture (client-server, microservices, IoT devices).
- **Project Management (e.g., Scrum/Agile).**

CHAPTER 4: Realization and Implementation (Practical)

- **Technological Choices**
- **Development Tools:** Hardware / and Software
- **Stack:** frontend/ backend.
- **Specialized APIs:** API for Geolocation...
- **System Architecture:**
 - Software Configuration
 - Hardware Setup
 - Connection & Communication
- **Implementation of Core Features (Logic + Screenshots):**
 - **Geolocation:** Code snippets + Live Map Screenshots.
 - **Smart Notifications:** Geofencing logic + Alert Screenshots.
- **Actor Interfaces:** Organize screenshots by user type (Admin vs. User).
- **Testing & Results**
- **Database Implementation:** Tables, sample data, and schema design.

CHAPTER 6: Business Model Canvas

GENERAL CONCLUSION

Summarize key findings, contributions, challenges, and potential future work or improvements.